TECHNOTRONICS™

Innovating the Future

User Manual: Drive Sense V.1 400m Laser Scanning Device

Overview: Long-Range Laser Scanning & Receiving System

This system is a sophisticated, portable duo designed for high-precision environmental mapping and long-distance data transmission. By pairing a high-powered **905nm laser scanner** with a dedicated **remote receiver**, the setup allows for real-time environmental monitoring and data visualization across significant distances.

1. The Scanner: Precision Data Acquisition

The Scanner serves as the primary sensory unit of the system, engineered to perform high-fidelity spatial mapping in demanding environments. It functions as a self-contained LiDAR station, actively probing the surroundings with pulsed light to translate physical topography into digital datasets. Encased in a ruggedized carbon fiber composite, the unit is designed to maintain optical alignment even during rapid X-Y axis movement, ensuring that every data point captured is both accurate and repeatable.

Key Features

- **Advanced Optics:** Utilizing a 905nm laser module and collimating lenses, the device boasts an impressive **400m range**. A dual-axis hinge system allows for precise X-Y scanning patterns.

- **Environmental Awareness:** Beyond distance, it integrates an **Si7021 sensor** to track ambient temperature and humidity, providing context to the scan data.

- **User Interface:** A 2.8-inch TFT LCD provides immediate feedback, protected by a durable polycarbonate shield.

- **Massive Endurance:** Power is handled by a substantial **9000mAh unified power supply**, ensuring the laser and dual-axis motors have the current they need for extended field use.

Main Scanner dimensions and weight

1. Length: 16cm, Width:10cm and Height: 10cm

2. Wall thickness: 3mm

3. Material: ABS+ Carbon Fiber

4. Weight of the scanner: 439gm

2. The Receiver: Remote Monitoring & Alerts

The **Receiver** acts as the central command hub for the operator, providing a remote interface to visualize and manage incoming data streams. By decoupling the display from the scanning hardware, the system allows for safe, long-range observation without a physical tether. It handles complex telemetry in real-time, translating raw laser data into a readable format while maintaining a high-speed wireless link to ensure low-latency updates from the field.

Key Features

- **Visual & Audio Feedback:** It features an **ILI9341 TFT LCD** for clear data readout and an integrated speaker for audio alerts, which is critical for notifying the user of specific environmental triggers or successful data captures.

- **Independent Portability:** With its own **5000mAh Li-ion battery** and dedicated charging ports, the receiver is designed for long-term handheld operation in the field.

The Receiver dimension and weight

1. Length: 13cm, Width: 10cm and Height: 10cm
2. Wall thickness: 3mm
3. Material: ABS+ Carbon Fiber
4. Weight of the display: 364gm

Quick Start Guide: Scanning System

1. Set Up the Receiver (Listen) * Power On: Turn on the receiver first.

   Ready Mode: Wait for the screen to show "Searching" or "Ready."

   This ensures that no data packets are missed when the scanner starts.

2. Set Up the Scanner (Transmit) * Position: Place the scanner on a stable surface

Power On: Turn the scanner on.

Initialization: Allow it a few seconds to calibrate and begin its rotation.

3. Monitor Real-Time Data * Link Check: Confirm the receiver display is populating with data.

Adjustment: If the data is lagging, ensure a clear line of sight between both units.

4. Power Down * Sequence: Turn off the **Scanner** first, then the **Receiver**.

🛡 Safety & Operational Guidelines

- **Enclosure Integrity:** For optimal performance and to protect the internal X-Y axis mechanics, ensure the device is operated only when the **protective housing is fully closed**.
- **Charging Protocol:** To preserve the lifespan of the 9000mAh and 5000mAh battery cells, use only a **regulated power supply** that matches the specified voltage requirements.

- **Component Protection:** Ensure all ports are clear of debris before connecting charging cables to prevent short circuits or pin damage.

🛠 Maintenance & Care

- **Lens & Screen Care:** Use a soft, dry microfiber cloth to keep the **polycarbonate screen and lenses** free of dust and fingerprints. Clear optics are essential for maintaining the 400m scan accuracy.

- **Environmental Shielding:** Avoid exposing the device to **solvents, chemical cleaners, or harsh aerosols**, as these can cloud the lenses and degrade the outer casing.

- **Mechanical Handling:** While the device is rugged, it contains precision dual-axis hinges. **Avoid drops or heavy impacts** to ensure the laser remains perfectly calibrated for X-Y movement.

Need Assistance? If you encounter any technical difficulties or hardware issues, please reach technotonics.uk@gmail.com.

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